JOURNAL OF MATHEMATICAL ANALYSIS AND APPLICATIONS 178, 470–487 (1993)

A Remark on the Regularity of the Solutions of the p-Laplacian and Its Application to Their Finite Element Approximation

W. B. LIU* AND JOHN W. BARRETT

Department of Mathematics, Imperial College, London SW7 2BZ, United Kingdom

Submitted by Harlan W. Stech

Received October 15, 1991

In this paper, global H^2 regularity is established for the solutions of the p-Laplacian equation, $1 < p \leqslant 2$, and related equations on domains which are either convex or have C^2 boundaries. The results are used to derive some explicit error bounds for the finite element approximation of the p-Laplacian, which were not previously available because of the lack of regularity results. © 1993 Academic Press, Inc.

1. INTRODUCTION

Let Ω be a bounded open set in R^2 with a Lipschitz boundary $\partial\Omega$. Many mathematical models from physical processes have the following form (the p-Laplacian): $(1 < p < \infty)$

$$\begin{aligned} -\nabla \cdot (|\nabla u|^{p-2} \nabla u) &= f \qquad \text{in} \quad \Omega \subset R^2 \\ u|_{\partial\Omega} &= g|_{\partial\Omega}, \end{aligned} \tag{1.1}$$

where the data f and g will be specified later. Such models arise from fluid mechanics (see [2, 3, 7]), nonlinear diffusion (see [24]) and nonlinear elasticity (see [1]). We note that the case where $1 < p < 2$ is of the most interest for elastic-plastic models. The well-posedness of (1.1) has been proved by monotonicity methods (see [9] and [14]); there being a unique solution in $W^{1,p}(\Omega)$. There has also been a great deal of work on the regularity of the solution of (1.1). For sufficiently regular data, local $C^{1,\alpha}$ regularity is established in [10, 27 28], and global $C^{1,\alpha}$ regularity in [19]. Higher order local regularity results are established in [17] for the case

* The financial support of the first author by SERC under Grant GR/F81255 is gratefully acknowledged.

0022-247X/93 $5.00

Copyright © 1993 by Academic Press, Inc.
All rights of reproduction in any form reserved.

$f \equiv 0$. In [20] the curvature of the corresponding level lines is studied. Clearly it would be desirable to prove higher order global regularity results. Below we see how such results are required in order to prove explicit error bounds for the finite element approximation of (1.1).

Let Ω be a polygonal domain and S^h be the space of continuous piecewise linear functions based on a regular triangulation of Ω. The finite element approximation of (1.1) from S^h is then: find $u^h \in V_g^h \equiv \{v^h \in S^h : v^h|_{\partial\Omega} = g^h\}$, where g^h is an appropriate approximation of g on $\partial\Omega$, such that:

$$\int_\Omega |\nabla u^h|^{p-2} (\nabla u^h, \nabla v^h) = \int_\Omega f v^h \qquad \forall v \in V_0^h. \tag{1.2}$$

Throughout this paper $(\cdot, \cdot)$ denotes the R^2 inner product. The well-posedness of (1.2) is easily established, see [14]. It is important to prove error bounds for this approximation; see for example [4, 6–9, 14, 21, 29]. We summarise these results below.

In [6] and [29] the following abstract error bound is established: for any $v^h \in V_g^h$

$$\|u - u^h\|_{W^{1,p}(\Omega)} \leqslant \begin{cases} M[\|u - v^h\|_{W^{1,p}(\Omega)}]^{p/2} & p \leqslant 2 \\ M[\|u - v^h\|_{W^{1,p}(\Omega)}]^{2/p} & p \geqslant 2 \end{cases}, \tag{1.3}$$

where M depends on $\|v^h\|_{W^{1,p}(\Omega)}$ for $p > 2$. Therefore it follows from standard approximation theory, see p. 123 in [9], that if $u \in W^{2,p}(\Omega)$ then

$$\|u - u^h\|_{W^{1,p}(\Omega)} \leqslant \begin{cases} Mh^{p/2} & p \leqslant 2 \\ Mh^{2/p} & p \geqslant 2 \end{cases}. \tag{1.4}$$

Recently, in [4] and [21] the following optimal error bounds have been established: if $p < 2$ and $u \in C^{2,2/p-1}(\bar{\Omega}) \cap W^{3,1}(\Omega)$ or $u \in W^{3,p}(\Omega)$, then

$$\|u - u^h\|_{W^{1,p}(\Omega)} \leqslant Mh \tag{1.5a}$$

and if $2 < p$, $u \in H^2(\Omega) \cap W^{1,\infty}(\Omega)$ and $|f| \geqslant \delta > 0$ then

$$\|u - u^h\|_{W^{1,1}(\Omega)} \leqslant Mh. \tag{1.5b}$$

In addition it is proved that the error bound (1.4) for $p > 2$ holds under the weaker regularity requirement $u \in H^2(\Omega) \cap W^{1,\infty}(\Omega)$. This is useful since it is well known that u is rarely in $W^{2,p}(\Omega)$ even if f and g are very smooth when $p > 2$, see [14], and so that (1.3) yields convergence rates much poorer than $2/p$.

All the above error bounds depend on the regularity of u. However, there appear to be no existing regularity results in the literature that can be

applied here, since all the results have either a first order or local character. It is the purpose of this short note to establish that the solutions of (1.1) in the case $1 < p < 2$ belong to $H^2(\Omega)$ if the domain is either convex or has a C^2 boundary. Then one can deduce some explicit error bounds for the piecewise linear finite element approximation, (1.2), by modifying the results in [21] and [29].

In order to establish $H^2(\Omega)$ regularity for the solution of (1.1), it is useful to examine first the regularized equation ($0 < \varepsilon \leqslant 1$):

$$\begin{aligned} -\nabla \cdot ((\varepsilon + |\nabla u|^2)^{p/2-1} \nabla u) &= f \quad \text{in} \quad \Omega \subset R^2 \\ u|_{\partial\Omega} &= g|_{\partial\Omega}. \end{aligned} \tag{1.6}$$

We note that (1.6) reduces to (1.1) if $\varepsilon = 0$. The equation (1.6) is also often met in practical problems. Its well-posedness is easily established and the error bounds (1.3)–(1.5) can equally be applied to its finite element approximation (see [6, 21]). Its regularity is also well established. For example for $1 < p \leqslant 2$, if $f \in C^\alpha(\bar{\Omega})$, $g \in C^{2,\alpha}(\bar{\Omega})$ for an $\alpha \in (0, 1)$ and $\partial\Omega \in C^2$, then $u \in H^2(\Omega) \cap C^{1,\alpha}(\bar{\Omega})$ (see Theorem 12.5 in [13] and Theorem 6.4.1 in [18]). If further $\partial\Omega \in C^{2,\alpha}$, then $u \in C^{2,\alpha}(\bar{\Omega})$ (see p. 309 in [13]).

The plan of this paper is as follows. In Section 2 we establish the $H^2(\Omega)$ regularity of the solutions of (1.6) and (1.1) under the assumptions $1 < p \leqslant 2$, $f \in L^q(\Omega)$ $(q > 2)$, $g \in H^2(\Omega)$ and Ω has a C^2 boundary. We then extend this result to Ω being a bounded convex set when $g = 0$. Furthermore, we extend the above results to the solutions of a general class of quasilinear elliptic equations. In Section 3 some explicit error bounds are obtained for the piecewise linear finite element approximation of such equations, by applying the regularity results established in Section 2.

Throughout this paper, we adopt the standard notation $W^{m,q}(\Omega)$ for Sobolev spaces on Ω with norm $\|\cdot\|_{W^{m,q}(\Omega)}$ and semi-norm $|\cdot|_{W^{m,q}(\Omega)}$. We set $W_0^{1,q} \equiv \{v \in W^{1,q}(\Omega) : v|_{\partial\Omega} = 0\}$ and we denote $W^{m,2}(\Omega)$ by $H^m(\Omega)$ and $W_0^{1,2}(\Omega)$ by $H_0^1(\Omega)$. For definitions of domains with C^k boundaries see [13] or [15]. In addition C and M denote two general positive constants independent of h.

2. Regularity of the Solutions of the p-Laplacian and Related Equations

The weak formulation of (1.1) is:

(WP): Find $u \in V_g \equiv \{v \in W^{1,p}(\Omega) : v|_{\partial\Omega} = g\}$ such that

$$\int_\Omega |\nabla u|^{p-2} (\nabla u, \nabla v) = \int_\Omega fv \qquad \forall v \in V_0,$$

where $f \in (V_0)^*$ (the dual space of V_0) and $g \in W^{1,p}(\Omega)$. Associated with (WP) is the following minimization problem:

(MP): Find $u \in V_g$ such that

$$J_\Omega(u) = \operatorname*{Min}_{v \in V_g} J_\Omega(v),$$

where

$$J_\Omega(v) \equiv \int_\Omega \int_0^{|\nabla v|} t^{p-1}\, dt - \int_\Omega fv.$$

It is not difficult to see that J_Ω is well defined and continuous on V_g. Moreover, J_Ω is Gateaux differentiable with $J'_\Omega(u)(v) \equiv \int_\Omega |\nabla u|^{p-2}(\nabla u, \nabla v) - \int_\Omega fv$. In the following we will always assume that $f \in L^2(\Omega)$. A solution of (WP) will be called a weak solution of (1.1). The well-posedness of (WP) has been well established. Furthermore J_Ω is strictly convex and coercive on V_g so that J_Ω has a unique minimizer on V_g, which satisfies its Euler equation (WP). In other words the problems (WP) and (MP) are equivalent. Similar conclusions hold for the weak formulation and equivalent minimization problem of (1.6), (see [6]).

We now examine the regularity of the weak solution of (1.1). First of all let us see some examples.

EXAMPLE 2.1. Let $\Omega = (-1, 1)$. Let u be the solution of (WP) $(1 < p \leqslant 2)$ with $u(-1) = u(1) = 0$. One finds that $u' = \operatorname{sign}(z)\,|z|^{1/(p-1)}$, where $z(t) = C - \int_{-1}^{t} f(s)\, ds$, and

$$u'' = [|z|^{(2-p)/(p-1)} z']/(p-1).$$

It follows that $u \in W^{2,2}(\Omega)$ if $f \in L^2(\Omega)$. It is easy to see that if $f \in C^{0,2/p-1}(\bar{\Omega})$, then $z' \in C^{0,2/p-1}(\bar{\Omega})$. From the inequality

$$||x|^\alpha - |y|^\alpha| \leqslant |x-y|^\alpha, \qquad \text{for any} \quad 0 < \alpha < 1,$$

one further concludes that $u'' \in C^{0,2/p-1}(\bar{\Omega})$. Furthermore one finds that

$$u''' = [|z|^{(2-p)/(p-1)} z'']/(p-1) + (2-p)[\operatorname{sign}(z)\,|z|^{(3-2p)/(p-1)}(z')^2]/(p-1)^2.$$

Therefore if in addition $f \in W^{1,1}(\Omega)$ then $u''' \in L^1(\Omega)$, since $z \in W^{2,1}(\Omega)$ and $(3-2p)/(p-1) > -1$ yield that (see [4])

$$|z|^{(3-2p)/(p-1)}(z')^2 \in L^1(\Omega).$$

Thus $u \in W^{3,1}(\Omega) \cap C^{2,2/p-1}(\bar{\Omega})$ if $f \in W^{1,1}(\Omega) \cap C^{0,2/p-1}(\bar{\Omega})$. Therefore u^h converges to u at least at the rate $h^{p/2}$ in $W^{1,p}(\Omega)$ if $f \in L^2(\Omega)$ from (1.4) and at the optimal rate h in $W^{1,p}(\Omega)$ if $f \in W^{1,1}(\Omega) \cap C^{0,2/p-1}(\bar{\Omega})$ from (1.5a).

In the above example for $1 < p < 2$ we see that u is sufficiently regular if the data is. When $2 < p < \infty$, however, it is a different story.

EXAMPLE 2.2. Let $\Omega = (-1, 1)$. Let $f(x) = \rho > 0$ and $u(-1) = u(1) = 0$. Consider a uniform mesh with nodes $x_i = ih$ and $h = 1/n$. The solutions u and u^h of (WP) and $(\mathrm{WP})^h$ are symmetric about $x = 0$ so that we only need analyse the half interval. One finds that $u' = C|x|^{1/(p-1)}$ so that $u \in W^{2,s}(\Omega)$ with $s(p-2) < (p-1)$. Thus the best one can expect for all p is that $u \in W^{2,1}(\Omega)$. It follows that $(u^h)'|_{(x_i, x_{i+1})} = u'((x_i + x_{i+1})/2) = C|(x_i + x_{i+1})/2|^{1/(p-1)}$. Consider the case $p = 3$. Then we have that

$$\int_0^1 |(u-u^h)'|^3 \, dx$$

$$= C \sum_{i=0}^{n-1} \int_{ih}^{(i+1)h} |x - (2i+1)h/2|^3 / |x^{1/2} + [(2i+1)h/2]^{1/2}|^3 \, dx$$

$$\geqslant C \sum_{i=0}^{n-1} \int_{ih}^{(i+1)h} |x - (2i+1)h/2|^3 \, dx / |[(i+1)h/2]^{1/2} + [(2i+1)h/2]^{1/2}|^3$$

$$\geqslant Ch^{5/2} \sum_{i=0}^{n-1} (1/(i+1))^{3/2} \geqslant Ch^{5/2},$$

since $\int_{i/n}^{(i+1)/n} |x - (2i+1)h/2|^3 \, dx = (h/2)^4/2$. Hence it follows that

$$\left[\int_0^1 |(u-u^h)'|^3 \, dx \right]^{1/3} \geqslant Ch^{5/6}.$$

Therefore, even if f and g are very smooth, it is possible that u is not in $W^{2,p}(\Omega)$, and $u^h \to u$ at a suboptimal rate in $W^{1,p}(\Omega)$. It is easy to extend the above analysis for this simple model problem to show that the convergence rate in $W^{1,p}(\Omega)$ is bounded below by $Ch^{(2p-1)/p(p-1)}$ for $p \geqslant (3 + \sqrt{5})/2$.

EXAMPLE 2.3. Consider the radially symmetric case of $\Omega = \{(r, \theta): r < 1\}$, $g = 0$ and $f(x, y) = F(r)$. It follows that

$$u(x, y) = U(r) = -\int_r^1 \operatorname{sign}[Z(t)] \, |Z(t)|^{1/(p-1)} \, dt,$$

where

$$Z(r) = -\int_0^r sF(s)\,ds/r.$$

One can prove that $u \in W^{2,2}(\Omega)$ if $f \in L^q(\Omega)$ with $q > 2$, $u \in C^{2,2/p-1}(\bar{\Omega})$ if $f \in C^{0,2/p-1}(\bar{\Omega})$ and $u \in W^{3,1}(\Omega)$ if $f \in W^{1,q}(\Omega)$ with $q > 1$, when $1 < p \leqslant 2$ (see [4]). Thus one obtains the desirable regularity in this case. On the other hand, if $p > 2$ and $f = \rho > 0$, then it is easy to see that $u \in W^{2,s}(\Omega)$ with $s(p-2) < 2(p-1)$ (see [4]). Therefore the best one can expect for all p is that $u \in W^{2,2}(\Omega)$.

We will mainly deal with the case $1 < p \leqslant 2$ and in the rest of the paper we will suppose this, unless otherwise stated. We now give our first regularity result:

THEOREM 2.1. *Let Ω be a bounded open set in R^2 with a C^2 boundary $\partial\Omega$. Let $g \in H^2(\Omega)$, $f \in L^q(\Omega)$ $(q > 2)$ and u be the unique weak solution of* (1.1). *Then $u \in H^2(\Omega)$ if $1 < p \leqslant 2$.*

Proof. If $p = 2$ the result is well known. Thus we will suppose that $p < 2$. Let $0 < \varepsilon \leqslant 1$ and $0 < \alpha < 1$. Let $f_\varepsilon \in C^\alpha(\bar{\Omega})$ and $g_\varepsilon \in C^{2,\alpha}(\bar{\Omega})$ be such that $f_\varepsilon \to f$ in $L^q(\Omega)$ and $g_\varepsilon \to g$ in $H^2(\Omega)$ as $\varepsilon \to 0$. Consider first the following equation

$$\begin{aligned} &[1 + (p-2)(v_1/v)^2]\,\partial^2 u_\varepsilon/\partial x_1^2 + [1 + (p-2)(v_2/v)^2]\,\partial^2 u_\varepsilon/\partial x_2^2 \\ &\quad + 2(p-2)(v_1 v_2/v^2)\,\partial^2 u_\varepsilon/\partial x_1\,\partial x_2 = -v^{2-p} f_\varepsilon \text{ in } \Omega \\ &\qquad u_\varepsilon|_{\partial\Omega} = g_\varepsilon|_{\partial\Omega}, \end{aligned} \tag{2.1}$$

where $v = (\varepsilon + |\nabla u_\varepsilon|^2)^{1/2}$, $v_1 = \partial u_\varepsilon/\partial x_1$ and $v_2 = \partial u_\varepsilon/\partial x_2$. It is easy to see that (2.1) is nothing other than (1.6) with data f_ε and g_ε. We have that (2.1) is a uniformly elliptic equation with $L^\infty(\Omega)$ coefficients, since for any $\xi \in R^2$

$$(p-1)\,|\xi|^2 \leqslant |\xi|^2 + (p-2)[\xi_1 v_1/v + \xi_2 v_2/v]^2 \leqslant |\xi|^2.$$

From Theorem 12.5 in [13] and Theorem 6.4.1 in [18] the nonlinear equation (2.1), for any $\varepsilon > 0$, has a unique solution $u_\varepsilon \in H^2(\Omega) \cap C^{1,\alpha}(\bar{\Omega})$. From the global H^2 estimate for linear elliptic equations with $L^\infty(\Omega)$ coefficients in two variables (see Theorem 37, III in [23]) we have that

$$\|u_\varepsilon\|_{H^2(\Omega)} \leqslant C(\|f_\varepsilon\,|\nabla u_\varepsilon|^{2-p}\|_{L^2(\Omega)} + \|g_\varepsilon\|_{H^2(\Omega)}), \tag{2.2}$$

where C is independent of ε. From a Holder inequality, Poincaré inequality and Sobolev imbedding theorems it follows that

$$\begin{aligned} \|f_\varepsilon\,|\nabla u_\varepsilon|^{2-p}\|_{L^2(\Omega)} &\leqslant C\,\|f_\varepsilon\|_{L^{q^*}(\Omega)}\,(\|u_\varepsilon\|_{W^{1,2(2-p)q^*/(q^*-2)}(\Omega)})^{2-p} \\ &\leqslant C(\|u_\varepsilon\|_{H^2(\Omega)})^{2-p}, \end{aligned}$$

where C is independent of ε and $2 < q^* \leqslant q$ is such that $2(2-p)q^*/(q^*-2) > 1$. Therefore $\{\|u_\varepsilon\|_{H^2(\Omega)}\}$ is bounded as $\varepsilon \to 0$ since $2-p<1$. Let $u_\varepsilon \to u$ strongly in $H^1(\Omega)$ and weakly in $H^2(\Omega)$. It is clear that $u \in V_g$. From (2.1) we have that

$$\int_\Omega (\varepsilon + |\nabla u_\varepsilon|^2)^{p/2-1} (\nabla u_\varepsilon, \nabla v) = \int_\Omega f_\varepsilon v \qquad \forall v \in V_0. \tag{2.3}$$

It is a simple matter to show that

$$|(\varepsilon + |\nabla u_\varepsilon|^2)^{p/2-1} \nabla u_\varepsilon - (\varepsilon + |\nabla u|^2)^{p/2-1} \nabla u| \leqslant M |\nabla(u_\varepsilon - u)|^{p-1}, \tag{2.4}$$

where M is independent of ε; see for example the proof of Proposition 3.2 in [3]. Letting $\varepsilon \to 0$, we obtain from (2.2)–(2.4) that

$$\int_\Omega |\nabla u|^{p-2} (\nabla u, \nabla v) = \int_\Omega fv \qquad \forall v \in V_0. \tag{2.5}$$

Therefore $u \in H^2(\Omega) \cap V_g$ solves (WP). ∎

Remark 2.1. It is known that the solution of (1.1) $(1 < p < \infty)$ belongs to $C^{1,\alpha}(\bar{\Omega})$ for an $\alpha > 0$ if $f \in L^\infty(\Omega)$, $g = 0$ and $\partial\Omega \in C^\infty$ (see [27]). From the proof of Proposition 1.1 in [16], it follows that $\partial\Omega \in C^2$ is sufficient to obtain this regularity. In addition we note that it may be possible to deduce the global H^2 regularity from various results in [27] and [28] if $f \in L^\infty(\Omega)$ and $g = 0$.

Finally, we note that it is obvious from the above proof that Theorem 2.1 is also true for the solution of (1.6). This relaxes the previously stated regularity requirements on the data for global H^2 regularity, see Section 1.

Remark 2.2. If $p > 2$ and $f = 0$, then it is easy to see from the proof that Theorem 2.1 still applies to (1.1) and obviously (1.6). In general, however, we do not know if Theorem 2.1 is true or not when $p > 2$. From Example 2.2 one sees that it is not true for the one dimensional case, but from Example 2.3 it may be true in higher dimensions.

In the following we would like to weaken the condition on $\partial\Omega$, which is quite restrictive in applications. In [15] it is shown that, similar to the linear case, the solutions of a class of quasilinear elliptic equations have H^2 regularity on bounded convex domains. This result, of course, is of great interest since in many practical problems the domain does not have a C^2 boundary but is a convex polygon. The equations examined in pp. 345–356 of [15] are of the form $-\nabla \cdot (\varphi(|\nabla u|)\nabla u) = f$, where φ is a positive non-decreasing function on $[0, \infty)$. In addition it is assumed that $a(t) = \varphi(t^{1/2})$ for $t \geqslant 0$ is a C^2 function. In the proof this differentiability condition on

$a(\cdot)$ plays an essential role. We note that this is not true for our problem as $a(t)=t^{(p-2)/2}$ with $1<p\leqslant 2$. Nevertheless, we can show a similar conclusion.

THEOREM 2.2. *Let Ω be a bounded open convex set in R^2. Let $g=0$, $f\in L^q(\Omega)$ $(q>2)$ and u be the unique weak solution of* (1.1). *Then $u\in H^2(\Omega)$ if $1<p\leqslant 2$.*

Proof. As before we can assume that $p<2$. We will follow the approach used in the proof of Theorem 3.2.1.2 in [15]. Let $\{\Omega_m\}$ $(m=1, 2, ...)$ be such that $\Omega_m\subseteq\Omega$, $\partial\Omega_m\in C^2$, Ω_m convex and $\operatorname{dist}(\partial\Omega, \partial\Omega_m)\to 0$ as $m\to\infty$. We know from Theorem 2.1 that the unique solution u_m of

$$-\nabla\cdot(|\nabla u_m|^{p-2}\nabla u_m)=f \text{ in } \Omega_m \qquad \text{and} \qquad u_m|_{\partial\Omega_m}=0, \tag{2.6}$$

is such that $u_m\in H^2(\Omega_m)\cap H_0^1(\Omega_m)$. Applying the approach of p. 173 in [23], we have that

$$|u_m|_{H^2(\Omega_m)}\leqslant C\,\| f|\nabla u_m|^{2-p}\|_{L^2(\Omega_m)}, \tag{2.7}$$

where C is independent of m because Ω_m is a convex set. Furthermore, one has that for any $\varepsilon>0$

$$\begin{aligned}\int_{\Omega_m}|\nabla u_m|^2 &= -\int_{\Omega_m} u_m\,\Delta u_m\leqslant(\varepsilon/2)\int_{\Omega_m}|u_m|^2+\int_{\Omega_m}|\Delta u_m|^2/2\varepsilon\\ &\leqslant C(\varepsilon/2)\operatorname{mes}(\Omega_m)\,|u_m|^2_{H^1(\Omega_m)}+(1/2\varepsilon)\,|u_m|^2_{H^2(\Omega_m)},\end{aligned}$$

where C is independent of m as $u_m\in H_0^1(\Omega_m)$, see for example p. 46 in [18]. By choosing ε sufficiently small it then follows, using the above Poincaré inequality again, that $\|u_m\|_{H^1(\Omega_m)}\leqslant C\,|u_m|_{H^2(\Omega_m)}$ and hence from (2.7) that

$$\|u_m\|_{H^2(\Omega_m)}\leqslant C\,\| f|\nabla u_m|^{2-p}\|_{L^2(\Omega_m)}, \tag{2.8}$$

where C is independent of m. In addition we note that

$$\| f|\nabla u_m|^{2-p}\|_{L^2(\Omega_m)}\leqslant M(\|u_m\|_{W^{1,2(2-p)q^*/(q^*-2)}(\Omega_m)})^{2-p},$$

where $2<q^*\leqslant q$ is such that $2(2-p)\,q^*/(q^*-2)>2$. As $u_m\in H^2(\Omega_m)\cap H_0^1(\Omega_m)$ we can apply the representation in Theorem 1.6.2 in [22] with $n=2$, $k=1$ and $l=2$ yielding

$$Du_m(x)=\sum_{|\beta|=2}\int_{\Omega_m}K_\beta(x-y)\,D^\beta u_m(y)\,|x-y|^{-1}\,dy,$$

where $|K_\beta| \leqslant C$ with C depending only on n, k and l. From the well known estimates for the Riesz potential (see, for example, Lemma 7.12 in [13]) it follows that

$$\|u_m\|_{W^{1,2(2-p)q^*/(q^*-2)}(\Omega_m)} \leqslant C_m \|u_m\|_{H^2(\Omega_m)},$$

where C_m only depends on the mes(Ω_m). One can reach the same conclusion by applying the imbedding Theorem 1.6.3.2 in [22] with $m=1$, $l=2$, $p=2-\varepsilon$ and $\varepsilon \in (0, 1)$. Therefore from (2.8) and reasoning as in the proof of Theorem 2.1, we have that $\{\|u_m\|_{H^2(\Omega_m)}\}$ is bounded.

Now let $\underline{u}_m$ be the zero extension of u_m. Then $\{\|\underline{u}_m\|_{H^1(R^2)}\}$ is bounded so that one can suppose that there is a subsequence of $\{\underline{u}_m\}$ (to be denoted by $\{\underline{u}_{m_k}\}$) converging to $\underline{u}$ weakly in $H^1(R^2)$. One can further suppose that for any open set $G \Subset \Omega$ (that is $\bar{G} \subset \Omega$), $\underline{u}_{m_k} \to \underline{u}$ strongly in $H^1(G)$. Let $u \equiv \underline{u}|_\Omega$. Then one has that $u \in H_0^1(\Omega)$. To see this; we note that $\underline{u}_{m_k}(1-\chi_\Omega)=0$ for all k, where χ_Ω is the characteristic function of Ω, so that $\underline{u}(1-\chi_\Omega)=0$ since $\underline{u}_{m_k} \to \underline{u}$ strongly in $L^2(\Omega)$. Thus $u \in \tilde{H}^1(\Omega) \equiv \{v \in H^1(\Omega)\colon \tilde{v} \in H^1(R^2)$, where $\tilde{v} \equiv v$ on Ω and 0 otherwise$\}$, see p. 18 of [15], since $\tilde{u} \equiv \underline{u}$. From Corollaries 1.2.2.3, 1.4.4.5 and 1.5.1.6 in [15], it follows that $\tilde{H}^1(\Omega) \equiv H_0^1(\Omega)$.

Let $\varphi \in \mathscr{D}(\Omega)$ and let $\varphi(k)$ be such that the support of φ is in Ω_{m_k} for $k \geqslant \varphi(k)$. From (2.6) for $k \geqslant \varphi(k)$ we have that

$$\int_\Omega |\nabla \underline{u}_{m_k}|^{p-2} (\nabla \underline{u}_{m_k}, \nabla\varphi) = \int_{\Omega_{m_k}} |\nabla \underline{u}_{m_k}|^{p-2} (\nabla \underline{u}_{m_k}, \nabla\varphi) = \int_{\Omega_{m_k}} f\varphi.$$

If one can show that $\lim_{k\to\infty} \int_{\Omega_{m_k}} |\nabla \underline{u}_{m_k}|^{p-2} (\nabla \underline{u}_{m_k}, \nabla\varphi) = \int_\Omega |\nabla \underline{u}|^{p-2} (\nabla \underline{u}, \nabla\varphi)$, then one has proved that u solves (WP) since $\int_{\Omega_{m_k}} f\varphi \to \int_\Omega f\varphi$.

Let $G \Subset \Omega$. Then it is clear that there is $G(k)$ such that $G \subset \Omega_m$ for $k \geqslant G(k)$ since $\bigcup \Omega_{m_k}$ is an open cover of $\bar{G}$. So for $k \geqslant \max(G(k), \varphi(k))$ it follows that

$$\begin{aligned}
&\left| \int_\Omega |\nabla \underline{u}|^{p-2} (\nabla \underline{u}, \nabla\varphi) - \int_{\Omega_{m_k}} |\nabla \underline{u}_{m_k}|^{p-2} (\nabla \underline{u}_{m_k}, \nabla\varphi) \right| \\
&\quad \leqslant \left| \int_{\Omega \setminus G} |\nabla \underline{u}|^{p-2} (\nabla \underline{u}, \nabla\varphi) \right| \\
&\qquad + \left| \int_{\Omega_{m_k} \setminus G} |\nabla \underline{u}_{m_k}|^{p-2} (\nabla \underline{u}_{m_k}, \nabla\varphi) \right| \\
&\qquad + \left| \int_G (|\nabla \underline{u}|^{p-2} \nabla \underline{u} - |\nabla \underline{u}_{m_k}|^{p-2} \nabla \underline{u}_{m_k}, \nabla\varphi) \right| \\
&\quad = T_1^k + T_2^k + T_3^k.
\end{aligned}$$

It is well known, see for example Lemma 1 in [6], that

$$T_3^k \leqslant M \|\varphi\|_{W^{1,p(G)}} [\|\underline{u} - \underline{u}_{m_k}\|_{W^{1,p(G)}}]^{p-1}.$$

On the other hand, we have that

$$\begin{aligned} T_1^k + T_2^k &\leqslant M \left[\left| \int_{\Omega \setminus G} |\nabla \underline{u}|^{p-1} \, d\Omega \right| + \left| \int_{\Omega_{m_k} \setminus G} |\nabla \underline{u}_{m_k}|^{p-1} \, d\Omega \right| \right. \\ &\leqslant M \Big[\operatorname{mes}(\Omega \setminus G) [\|\underline{u}\|_{W^{1,p(\Omega)}}]^{p-1} \\ &\quad \left. + \operatorname{mes}(\Omega_{m_k} \setminus G) [\|\underline{u}_{m_k}\|_{W^{1,p(\Omega_{m_k})}}]^{p-1} \right]. \end{aligned}$$

For any $\varepsilon > 0$, we choose $G \Subset \Omega$ such that $T_1^k + T_2^k < \varepsilon/2$; this is possible since $\{\|\underline{u}_{m_k}\|_{W^{1,p(\Omega_{m_k})}}\}$ is bounded above. Then we take $k \geqslant \max(G(k), \varphi(k))$ such that $T_3^k < \varepsilon/2$. This yields that $T_1^k + T_2^k + T_3^k < \varepsilon$ for k sufficiently large. Therefore $\lim_{k \to \infty} \int_{\Omega_{m_k}} |\nabla \underline{u}_{m_k}|^{p-2} (\nabla \underline{u}_{m_k}, \nabla \varphi) = \int_\Omega |\nabla \underline{u}|^{p-2} (\nabla \underline{u}, \nabla \varphi)$. It follows that u is the unique solution of (1.1).

We now show that $u \in H^2(\Omega)$ by the same reasoning as used in the proof of Theorem 3.2.1.2 in [15]. Let $v_{m,i,j} \equiv (\partial^2 u_m / \partial x_i \, \partial x_j)^{\sim}$. It follows that $\{v_{m,i,j}\}_{m \geqslant 1}$ is bounded and hence there exists a subsequence $\{v_{m_k,i,j}\}_{k \geqslant 1}$ converging to $v_{i,j}$ weakly in $L^2(R^2)$. For $k \geqslant \varphi(k)$ we have that

$$\begin{aligned} \int_\Omega \underline{u}_{m_k} (\partial^2 \varphi / \partial x_i \, \partial x_j) &= \int_{\Omega_{m_k}} u_{m_k} (\partial^2 \varphi / \partial x_i \, \partial x_j) \\ &= \int_{\Omega_{m_k}} (\partial^2 u_{m_k} / \partial x_i \, \partial x_j) \varphi \\ &= \int_\Omega v_{m_k,i,j} \varphi. \end{aligned}$$

Letting $k \to \infty$, yields that $\int_\Omega u (\partial^2 \varphi / \partial x_i \, \partial x_j) = \int_\Omega v_{i,j} \varphi$ for $1 \leqslant i, j \leqslant 2$ and hence we have that $u \in H^2(\Omega)$. ∎

Remark 2.3. One can prove that $\{\|u_m\|_{H^2(\Omega_m)}\}$ is bounded in an alternative way to that in the above. As Ω is a bounded convex set, one can ensure that the Lipschitz constants of Ω_m, $m = 1, 2, \ldots$, are uniformly bounded (or the geometric constants θ, h and r in the domain class $\Pi(\theta, h, r)$, see [5], are uniformly bounded away from zero). This is obvious when Ω is a polygonal domain. When Ω is a general convex set, one can initially approximate it with a class of convex polygonal domains Ω_m^* whose Lipschitz constants are uniformly bounded. Then instead of using the result in [22], we apply Theorem II.1 in [5]: there is an

extension operator E_m from $H^2(\Omega_m)$ to $H^2(\Omega)$ for $m = 1, 2, \ldots$ such that $\{\|E_m\|\}$ is bounded. Thus we have that

$$\|u_m\|_{W^{1,2(2-p)q^*/(q^*-2)}(\Omega_m)} \leqslant \|E_m u_m\|_{W^{1,2(2-p)q^*/(q^*-2)}(\Omega)} \leqslant C\,\|E_m u_m\|_{H^2(\Omega)} \leqslant C\,\|u_m\|_{H^2(\Omega_m)},$$

where C is independent of m. Therefore it follows as before that the sequence $\{\|u_m\|_{H^2(\Omega_m)}\}$ is bounded.

From Theorem 2.2 we can prove another interesting result:

THEOREM 2.3. *Let Ω be a bounded open convex set in R^2 with a polygonal boundary. Let $g = 0$, $f \in L^q(\Omega)$ $(q > 2)$ and u be the unique weak solution of* (1.1). *Then $u \in C^{1,\alpha}(\bar{\Omega})$ for an $\alpha \in (0, 1)$ if $1 < p \leqslant 2$.*

Proof. From Theorem 2.2 we know that there is a unique solution u in $H^2(\Omega) \cap H_0^1(\Omega)$ to (WP) such that

$$[1 + (p-2)(v_1/v)^2]\,\partial^2 u/\partial x_1^2 + [1 + (p-2)(v_2/v)^2]\,\partial^2 u/\partial x_2^2 + 2(p-2)v_1 v_2/v^2\,\partial^2 u/\partial x_1\,\partial x_2 = -v^{2-p} f \quad \text{in } \Omega,$$

with $v = |\nabla v|$, $v_1 = \partial u/\partial x_1$ and $v_2 = \partial u/\partial x_2$. Therefore all the coefficients in the above equation are bounded and it follows that it is uniformly elliptic. Furthermore $f\,|\nabla u|^{2-p} \in L^{q'}(\Omega)$ with $q' > 2$. From Corollary 8.1.6 in [15] it follows that $\nabla u \in C^{0,\alpha}(\bar{\Omega})^2$ for an $\alpha \in (0, 1)$. On the other hand $u \in H^2(\Omega)$ so the desired result follows. ∎

Remark 2.4. We note that it does not appear possible to prove Theorem 2.3 by the approach used in [27] as $\partial\Omega$ is not smooth. From the proofs of Theorems 2.2 and 2.3 it is easy to see that the conclusions are also true for (1.6). The results in [15] still cannot be applied to (1.6) as the function $t \to (\varepsilon + t^2)^{p/2-1}$ $(1 < p < 2)$ is a decreasing function. We would like to know whether Theorems 2.2 and 2.3 are true or not in the case $p > 2$, see Remark 2.2. Once again the theory in [15] does not apply as $a(0) = 0$.

Remark 2.5. Up to now we have only dealt with the Dirichlet problem. Since the H^2 estimates for uniformly elliptic linear equations with $L^\infty(\Omega)$ coefficients in two variables also hold for more general boundary conditions on a convex domain (see p. 228 in [18]), Theorem 2.1 can be extended to this class of problems and the proofs are basically the same. Theorem 2.2 can also be extended by using techniques in Section 3.2 in [15]. Even if Ω is not convex, we can still extend Theorem 2.1 to mixed boundary conditions. To do this we note that (3.17.8) in [18] is still true and that J_1 in (3.17.9) can still be transformed into integrals over $\partial\Omega$ that

do not contain derivatives of u higher than the first. Then by using (37.6) in [23] and the same techniques as employed in the proof of Theorem 37, III in [23], one can see that the corresponding version of (3.17.5) in [18] is still true for a C^2 domain and mixed boundary conditions. The remainder of the proof follows that for convex domains.

Remark 2.6. Although the H^2 estimates used above are true only in two dimensions, we are still able to extend some of our results to three dimensions. Consider the three dimensional version of Theorem 2.1 under the modified assumptions: $\partial\Omega \in C^3$ and has positive mean curvature, $g \in H^2(\Omega)$ and $f \in L^q(\Omega)$ $(q > 6/(1+p))$ with $1 < p \leqslant 2$. We first note from Theorem 15.12 in [13] that if $f \in C^\alpha(\bar{\Omega})$ and $g \in C^{2,\alpha}(\bar{\Omega})$ for an $\alpha \in (0, 1)$, then the solution of (1.6) belongs to $C^{2,\alpha}(\bar{\Omega})$. We note next that for any fixed $p > 1$, there is $\delta(p) > 0$ such that (2.1) satisfies the inequality (37.14) in [23]. It follows that the estimate (37.13) in [23] holds and hence (2.2) holds with C independent of ε. Then noting that $q^* > 6/(1+p)$ yields that $r = 2(2-p)q^*/(q^*-2) < 6$ and hence $H^2(\Omega) \subsetneq W^{1,r}(\Omega)$ we obtain the desired result by following the proof of Theorem 2.1 above.

From the above we can prove a three dimensional version of Theorem 2.2 for any three dimensional convex domain which can be approximated by C^3 domains with positive mean curvature, for example a cube, provided $g = 0$ and $f \in L^q(\Omega)$ $(q > 6/(1+p))$ with $1 < p \leqslant 2$. The proof is virtually the same as Theorem 2.2. The only point to note is that the constant C in the corresponding versions of (2.7) and (2.8) are independent of m by inspecting [25], where the proof of (37.13) in [23] is given. As for Theorem 2.3 we do not know if it still holds in three dimensions.

We now return to two dimensional problems and examine a more general equation. For a positive function k such that $t \to k(t)t$ belongs to $C^0[0, \infty) \cap C^1(0, \infty)$ consider the following quasilinear elliptic equation:

$$-\nabla \cdot (k(|\nabla u|)\nabla u) = f \qquad \text{in } \Omega \subset R^2 \qquad \text{and} \qquad u|_{\partial\Omega} = g|_{\partial\Omega}, \quad (2.9)$$

where the data f and g satisfy the same assumptions as in Theorem 2.1. In [6, 21] its well-posedness has been established for a class of functions k. For example, (2.9) has a unique weak solution in V_g if there is a $p \in (1, 2]$ and $M > 0$ such that for any $s, t \geqslant 0$

$$|k(t)t - k(s)s| \leqslant M\,|t-s|^{p-1} \quad (2.10a)$$

and

$$(1 + t^{2-p})\, d(k(t)t)/dt \geqslant 1/M. \quad (2.10b)$$

The error bounds (1.3) and (1.4) also apply to the finite element approximation of (2.9), ((3.1)). However for (1.5a) to hold, (2.10a) must be replaced by an alternative condition (see [21])). Associated with (2.9) is the following minimization problem: Find $u \in V_g$ such that

$$J_\Omega(u) = \operatorname*{Min}_{v \in V_g} J_\Omega(v), \tag{2.11a}$$

where

$$J_\Omega(v) \equiv \int_\Omega \int_0^{|\nabla v|} k(t)\, t\, dt - \int_\Omega fv. \tag{2.11b}$$

It is well known that J_Ω is continuous and Gateaux differentiable on V_g with $J'_\Omega(u)(v) \equiv \int_\Omega k(|\nabla u|)(\nabla u, \nabla v) - \int_\Omega fv$. Further one has that J_Ω is strictly convex and coercive on V_g so that J_Ω has a unique minimizer on V_g, which satisfies its Euler equation (2.9) (see [6]). Therefore (2.9) is equivalent to (2.11). In the following we require that k satisfies the additional condition

$$|k'(t)t/k(t)| \leqslant 2 - p \qquad \text{for} \quad t \geqslant 0. \tag{2.12}$$

We note that the conditions (2.10) and (2.12) can infer (1.2)–(1.5) in [26]. Some examples of $k(t)$ satisfying these conditions are t^{p-2}, $(\varepsilon + t^2)^{p/2-1}$ and $[t(1+t)]^{p/2-1}$ with $1 < p \leqslant 2$.

THEOREM 2.4. *Assume that* (2.10) *and* (2.12) *hold. Let* $g \in H^2(\Omega)$, $f \in L^q(\Omega)$ $(q > 2)$ *and* u *be the unique weak solution of* (2.9).

(i) *If* $\partial\Omega \in C^2$ *then* $u \in H^2(\Omega)$.

(ii) *If* Ω *is convex and* $g = 0$, *then* $u \in H^2(\Omega)$.

(iii) *If* Ω *is convex with a polygonal boundary and* $g = 0$, *then* $u \in C^{1,\alpha}(\bar{\Omega})$ *for an* $\alpha \in (0, 1)$.

Proof. The proofs of (i)–(iii) are very similar to those given in Theorems 2.1–2.3 and so only outlines will be given here. Following an idea in [26] let $k_\varepsilon(t) = k((\varepsilon + t^2)^{1/2})$, which is well defined and differentiable on $[0, \infty)$. Consider the following equation:

$$-\nabla \cdot (k_\varepsilon(|\nabla u_\varepsilon|)\nabla u_\varepsilon) = f_\varepsilon \qquad \text{in } \Omega \qquad \text{and} \qquad u_\varepsilon|_{\partial\Omega} = g_\varepsilon|_{\partial\Omega}, \tag{2.13}$$

where f_ε, g_ε are sufficiently smooth and converging to f, g as before.

(i) In this case, the unique solution of (2.13) $u_\varepsilon \in H^2(\Omega)$ for any $\varepsilon > 0$ (see [13, 18, or 25]) or alternatively this result follows from noting that (2.13) can be written as

$$
\begin{aligned}
&[1 + [k'(v)\, v/k(v)](v_1/v)^2]\, \partial^2 u_\varepsilon/\partial x_1^2 \\
&\quad + [1 + [k'(v)\, v/k(v)](v_2/v)^2]\, \partial^2 u_\varepsilon/\partial x_2^2 \\
&\quad + 2[k'(v)\, v/k(v)](v_1 v_2/v^2)\, \partial^2 u_\varepsilon/\partial x_1\, \partial x_2 \\
&= -k(v)^{-1} f_\varepsilon \qquad \text{in } \Omega, \\
&\qquad u_\varepsilon|_{\partial\Omega} = g_\varepsilon|_{\partial\Omega},
\end{aligned}
$$

where $v = (\varepsilon + |\nabla u_\varepsilon|^2)^{1/2}$, $v_1 = \partial u_\varepsilon/\partial x_1$ and $v_2 = \partial u_\varepsilon/\partial x_2$. From (2.10b) it follows that $k(v)^{-1} \leqslant M(1 + v^{2-p})$, see [6]. Noting this and (2.12) it follows that this equation is uniformly elliptic with bounded coefficients and $\{\|u_\varepsilon\|_{H^2(\Omega)}\}$ is bounded as in the proof of Theorem 2.1. Let $u_\varepsilon \to u^* \in H^2(\Omega)$ strongly in $H^1(\Omega)$. On the other hand, $u_\varepsilon - \hat{u} \to 0$ in $W^{1,p}(\Omega)$, where $\hat{u}_\varepsilon$ is the solution (2.9) with $f = f_\varepsilon$, see p. 731 in [26]. Furthermore $\hat{u}_\varepsilon \to u$ in $W^{1,p}(\Omega)$, where u is the solution of (2.9), see p. 385 in [6]. Therefore we have that $u^* \equiv u \in H^2(\Omega)$.

(ii) We follow the same approach as used in the proof of Theorem 2.2. We have that $\{\|u_m\|_{H^2(\Omega_m)}\}$ is bounded. Let $\underline{u}_m \to u$ weakly in $H^1(\Omega)$, where $\underline{u}_m$ is the zero extension of u_m. From (2.10a) it follows that $|k(|\nabla v|)\, \nabla v| \leqslant M\, |\nabla v|^{p-1}$ and that

$$
\begin{aligned}
&\left| \int_G (k(|\nabla u|)\, \nabla u - k(|\nabla u_m|)\, \nabla u_m, \nabla \varphi) \right| \\
&\qquad \leqslant M\, \|\varphi\|_{W^{1,p}(G)} [\|u - u_m\|_{W^{1,p}(G)}]^{p-1},
\end{aligned}
$$

see [6]. Therefore the desired result follows.

(iii) The same argument as used in the proof of Theorem 2.3 can be applied. ∎

3. Finite Element Approximation and Abstract Error Bounds

In this section we consider the piecewise linear finite element approximation of (2.9) and give some explicit error bounds by using the regularity results derived in the previous section. We assume that $f \in L^q(\Omega)$ $(q > 2)$ and for ease of exposition that $g = 0$.

Let Ω^h be a polygonal approximation to Ω with boundary $\partial\Omega^h$. Let T^h be a partitioning of Ω^h into disjoint open regular triangles τ, each of maximum diameter bounded above h, so that $\bar{\Omega}^h = \bigcup_{\tau \in T^h} \bar{\tau}$. We assume that $P_i \in \partial\Omega^h \Rightarrow P_i \in \partial\Omega$, where $\{P_i\}$ $(i = 1 \cdots I)$ is the vertex set associated with the triangulation T^h. For ease of exposition we assume that $\Omega^h \subseteq \Omega$.

Associated with T^h is a finite dimensional subspace S^h of $C^0(\bar{\Omega}^h)$, such that $v^h|_\tau$ is linear for all $v^h \in S^h$ and $\tau \in T^h$. Let

$$V_0^h = \{v^h \in S^h : v^h(P_i) = 0 \text{ for any } P_i \in \partial\Omega\},$$

and so $V_0^h \subset W_0^{1,p}(\Omega^h)$. Let $\pi_h : C(\bar{\Omega}^h) \to S^h$, denote the interpolation operator such that for any $v \in C(\bar{\Omega}^h)$, $\pi_h v(P_i) = v(P_i)$ $i = 1 \to I$.

A finite element approximation of (2.9) is then: Find $u^h \in V_0^h$ such that

$$\int_{\Omega^h} k(|\nabla u^h|)(\nabla u^h, \nabla v^h) = \int_{\Omega^h} f v^h, \qquad \forall v^h \in V_0^h. \tag{3.1}$$

Equivalently, $u^h \in V_0^h$ solves the minimization problem

$$J_{\Omega^h}(u^h) = \operatorname*{Min}_{v^h \in V_0^h} J_{\Omega^h}(v), \tag{3.2}$$

where $J_{\Omega^h}(\cdot)$ is given by (2.11) with Ω replaced by Ω^h. The well-posedness of the above is also well-established under the assumptions (2.10), see [6]. Below we modify the abstract error bound in [6], see [21].

THEOREM 3.1. *Assume that Ω is a convex domain, k satisfies* (2.10) *and* (2.12), $g = 0$ *and* $f \in L^q(\Omega)$ $(q > 2)$. *Then u and u^h, the solutions of* (2.9) *and* (3.2), *are such that*

$$\|u - u^h\|_{W^{1,p}(\Omega^h)} \leqslant Ch^{p/2}. \tag{3.3a}$$

If in addition the function $t \to k(t)t$ is Lipschitz, then

$$\|u - u^h\|_{W^{1,p}(\Omega^h)} \leqslant Ch. \tag{3.3b}$$

Proof. From (ii) of Theorem 2.4 we have that $u \in H^2(\Omega)$. As Ω is convex we have that $\Omega^h \subseteq \Omega$. Therefore in the case $\Omega^h \equiv \Omega$ the result (3.3a) follows directly from choosing $v^h \equiv \pi_h u$ in (1.3) and applying the standard approximation results, see p. 123 in [9]. It is a simple matter to adapt the proof of (1.3) for $p < 2$ given in Section 7 of [6] to the case $\Omega^h \subseteq \Omega$. We obtain that for any $v^h \in V_0^h$

$$A(u^h) \leqslant A(v^h) + J'_{\Omega^h}(u)(v^h - u^h),$$

where

$$A(v^h) \equiv \int_0^1 \int_{\Omega^h} (k(|\nabla(u + s(v^h - u))|)\nabla(u + s(v^h - u)) - k(|\nabla u|)\nabla u, \nabla(v^h - u))\, ds.$$

As $-\nabla.(k(|\nabla u|)\nabla u) = f \in L^2(\Omega)$ on Ω and $\Omega \subseteq \Omega^h$, it follows that $J'_{\Omega^h}(u)(v^h - u^h) = 0$ and hence

$$A(u^h) \leqslant A(v^h) \qquad \forall v^h \in V_0^h. \tag{3.4}$$

From [6] it follows that

$$\begin{aligned} C \|u - u^h\|^2_{W^{1,p}(\Omega^h)} &\leqslant A(u^h) \leqslant A(v^h) \\ &\leqslant M \|u - v^h\|^p_{W^{1,p}(\Omega^h)} \qquad \forall v^h \in V_0^h, \end{aligned}$$

which immediately yields the generalisation of (1.3) for $p < 2$ to the case $\Omega^h \subseteq \Omega$. The desired result (3.3a) then follows as in the case $\Omega^h \equiv \Omega$ given above.

When $t \to k(t)t$ is Lipschitz, it follows that

$$A(v^h) \leqslant M \|u - v^h\|^2_{W^{1,2}(\Omega^h)} \qquad \forall v^h \in V_0^h,$$

since $u \in H^2(\Omega)$. Thus we obtain (3.3b) by taking $v^h = \pi_h u$. ∎

Remark 3.1. As an application of the Theorem 3.1 we have for $1 < p \leqslant 2$ that (3.3a) holds if $k(t) = t^{p-2}$ and (3.3b) holds if $k(t) = (\varepsilon + t^2)^{p/2-1}$. In addition it is easily seen from Remark 2.6 that Theorem 3.1 generalises to convex domains in three dimensions if $f \in L^q(\Omega)$ $(q > 6/(p+1))$ with S^h being the continuous piecewise linear functions on a partitioning of Ω^h into regular tetrahedra.

The requirement $\Omega^h \subseteq \Omega$ is somewhat restrictive in practice. In the following we remove this restriction when k is smooth.

THEOREM 3.2. *Assume that* $\partial\Omega \in C^{2,\alpha}$, $k \in C^{1,\alpha}[0, \infty)$ and $f \in C^\alpha(\bar{\Omega})$ for an $\alpha \in (0, 1)$; *and* k *satisfies* (2.8) *and* (2.10). *Let* u *be the solution of* (2.7) *and* u^h *be the solution of* (3.1) *with* $f = \tilde{f}$, *where* $\tilde{f} \equiv f$ *on* Ω *and* $\tilde{f} \equiv 0$ *otherwise. Then we have that*

$$\|u - u^h\|_{W^{1,p}(\Omega \cap \Omega^h)} \leqslant Ch. \tag{3.5}$$

Proof. It follows from Theorem 12.5 and p. 309 in [13] that $u \in C^{2,\alpha}(\bar{\Omega})$. Then there exists an extension $\hat{u} \in C^{2,\alpha}(\overline{\Omega \cup \Omega^h})$ such that $\hat{u} = u$ on Ω, see for example Lemma 6.37 in [13]. Let

$$\hat{f} = \begin{cases} f & \text{if } x \in \Omega, \\ -\nabla \cdot (k(|\nabla \hat{u}|)\nabla \hat{u}) & \text{otherwise.} \end{cases} \tag{3.6}$$

It follows that $\hat{f} \in L^{\infty}(\Omega \cup \Omega^h)$ since $\hat{u} \in C^{2,\alpha}(\overline{\Omega \cup \Omega^h})$. From the proof of Theorem 3.1 as $t \to k(t)t$ is Lipschitz we have that

$$\|\hat{u} - u^h\|^2_{W^{1,p}(\Omega^h)} \leqslant Ch^2 + \tilde{J}'_{\Omega^h}(\hat{u})(\pi_h \hat{u} - u^h),$$

where $\tilde{J}_{\Omega^h}$ is given by (2.11b) with Ω replaced by Ω^h and f replaced by $\tilde{f}$. From (3.6) it follows that

$$\begin{aligned}
&|\tilde{J}'_{\Omega^h}(\hat{u})(\pi_h \hat{u} - u^h)| \\
&\quad \leqslant \int_{\Omega^h \setminus (\Omega \cap \Omega^h)} |\tilde{f} - \hat{f}|\, |(\pi_h \hat{u} - u^h)| \\
&\quad \leqslant C \operatorname{mes}(\Omega^h \setminus (\Omega \cap \Omega^h))^{1/2} \left[\|\pi_h \hat{u} - \hat{u}\|_{L^2(\Omega^h)} + \|\hat{u} - u^h\|_{W^{1,p}(\Omega^h)} \right] \\
&\quad \leqslant Ch^2 + Ch\, \|\hat{u} - u^h\|_{W^{1,p}(\Omega^h)},
\end{aligned}$$

where we have noted that $\operatorname{dist}(\partial\Omega, \partial\Omega^h) \leqslant Ch^2$. Hence we obtain the desired result (3.5). ∎

Remark 3.2. To obtain an optimal error bound for the p-Laplacian for $1 < p < 2$, see (1.5a), one needs to prove that the solution $u \in C^{2,2/p-1}(\bar{\Omega}) \cap W^{3,1}(\Omega)$ or $W^{3,p}(\Omega)$. We will discuss such regularity in a forthcoming paper.

REFERENCES

1. C. ATKINSON AND C. R. CHAMPION, Some boundary value problems for the equation $\nabla.(|\nabla\phi|^N \nabla\phi) = 0$, *Quart. J. Mech. Appl. Math.* **37** (1984), 401–419.
2. C. ATKINSON AND C. W. JONES, Similarity solutions in some non-linear diffusion problems and in boundary-layer flow of a pseudo plastic fluid, *Quart. J. Mech. Appl. Math.* **27** (1974), 193–211.
3. J. BARANGER AND K. NAJIB, Numerical analysis for quasi-Newtonian flow obeying the power law or the Carreau law, *Numer. Math.* **58** (1990), 35–49.
4. J. W. BARRETT AND W. B. LIU, Finite element approximation of the p-Laplacian, *Math. Comp.*, to appear.
5. D. CHENAIS, On the existence of a solution in a domain identification problem, *J. Math. Anal. Appl.* **52** (1975), 189–219.
6. S. S. CHOW, Finite element error estimates for non-linear elliptic equations of monotone type, *Numer. Math.* **54** (1989), 373–393.
7. S. S. CHOW, "Finite Element Error Estimates for Non-linear Elliptic Equations of Monotone Type," Ph.D. Thesis, Aust. Nat. Univ., Canberra, Australia, 1983.
8. P. G. CIARLET, M. H. SCHULTZ, AND R. S. VARGA, Numerical methods of high order accuracy for nonlinear boundary value problems: V. Monotone operator theory, *Numer. Math.* **13** (1969), 51–77.
9. P. G. CIARLET, "The Finite Element Method for Elliptic Problems," North Holland, Amsterdam, 1978.

10. E. DiBENEDETTO, $C^{1+\alpha}$ local regularity of weak solutions of degenerate elliptic equations, *Nonlinear Anal.* **7** (1983), 827–850.
11. M. DOBROWOLSKI AND R. RANNACHER, Finite element methods for nonlinear elliptic systems of second order, *Math. Nachr.* **94** (1980), 155–172.
12. J. FREHSE AND R. RANNACHER, Asymptotic L^{∞}-error estimates for linear finite element approximations of quasilinear boundary problems, *SIAM. J. Numer. Anal.* **15** (1978), 419–431.
13. D. GILBARG AND N. TRUDINGER, "Elliptic Partial Differential Equations of Second Order," second ed., Springer-Verlag, Berlin, 1984.
14. R. GLOWINSKI AND A. MARROCCO, Sur l'approximation par éléments finis d'ordre un, et la résolution, par pénalisation-dualité, d'une classe de problèmes de Dirichlet non linéaires, *RAIRO Anal. Num.* **2** (1975), 41–64.
15. P. GRISVARD, "Elliptic Problems in Nonsmooth Domains," Pitman, Boston, 1985.
16. M. GUEDDA AND L. VERON, Quasilinear elliptic equations involving critical Sobolev exponents, *Nonlinear Anal.* **13** (1989), 879–902.
17. T. IWANIEC AND J. J. MANFREDI, Regularity of p-harmonic functions on the plane, *Rev. Math. Iberoamericana* **5** (1989), 1–19.
18. O. A. LADYZHENSKAYA AND N. N. URAL'TSEVA, "Linear and Quasilinear Elliptic Equations," Academic Press, New York, 1968.
19. G. M. LIEBERMAN, Boundary regularity for solutions of degenerate elliptic equations, *Nonlinear Anal.* **12** (1988), 1203–1219.
20. P. LINDQVIST, On p-harmonic functions in the complex plane and curvature, *Israel. J. Math.* **63** (1988), 257–269.
21. W. B. LIU AND J. W. BARRETT, Finite element approximation of some degenerate monotone quasilinear elliptic systems, submitted for publication.
22. V. G. MAZ'JA, "Sobolev Spaces," Springer-Verlag, Berlin, 1985.
23. C. MIRANDA, "Partial Differential Equations of Elliptic Type," second ed., Springer-Verlag, Berlin, 1970.
24. J. R. PHILIP, N-diffusion, *Aust. J. Phys.* **14** (1961), 1–13.
25. G. TALENTI, Sopra una classe di equazioni ellittiche a coefficienti misurabili, *Ann. Mat. Pura Appl.* **69** (1965), 285–304.
26. P. TOLKSDORF, On quasilinear boundary value problems in domains with corners, *Nonlinear Anal.* **5** (1981), 721–735.
27. P. TOLKSDORF, On the Dirichlet problem for quasilinear equations in domains with conical boundary points, *Comm. Partial. Differential Equations* **8** (1983), 773–817.
28. P. TOLKSDORF, Regularity for a more general class of quasilinear elliptic equations, *J. Differential Equations* **51** (1984), 126–150.
29. V. B. TYUKHTIN, The rate of convergence of approximation methods of one-sided problems I, *Vestnik Leningrad Univ. Mat. Mekh. Astronom.* **13** (1982), 111–113. [in Russian]